Fermi Estimation Fine-Tuning Pipeline

DS595 — Final Project

What We Targeted — and Why

A **Fermi estimation** is a back-of-the-envelope calcula
impossible quantity.

Named after Enrico Fermi — "How many piano tuners

Why it matters

- Engineering feasibility checks
- Market sizing in tech interviews
- Scientific sanity checks


Evaluat

Dataset

Master dataset: 5,008 questions

Integrated from four sources:

Source	Type
Science Olympiad (scioly)	Competition Qs
FermiP (realFP)	Real-world Qs
FermiQ (fermiq)	Physics/sci Qs
Student traces	Model-generated reasoning

Scoring & Prompt Contract

FP Score (lenient, eval)

```
score = max(0, 1 - δ / 3.0)
```

RL OOM Score (strict, training)

```
score = max(0, 1 - δ / 1.5)
```

where **$\delta = |\log_{10}(\text{pred}) - \text{gold OOM}|$**

Prompt Contract — Example

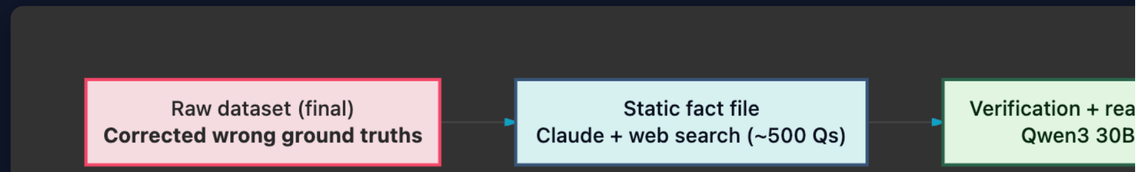
SYSTEM

```
You are a Fermi estimation solver. Decompose the problem
into a small number of factors, bound each factor with
low/high/point estimates from first principles (no
lookups, no search), combine them with an explicit
formula, and return a single numeric answer in the
requested unit.

Respond with ONE JSON object. No prose before or after. No
code fences.
```

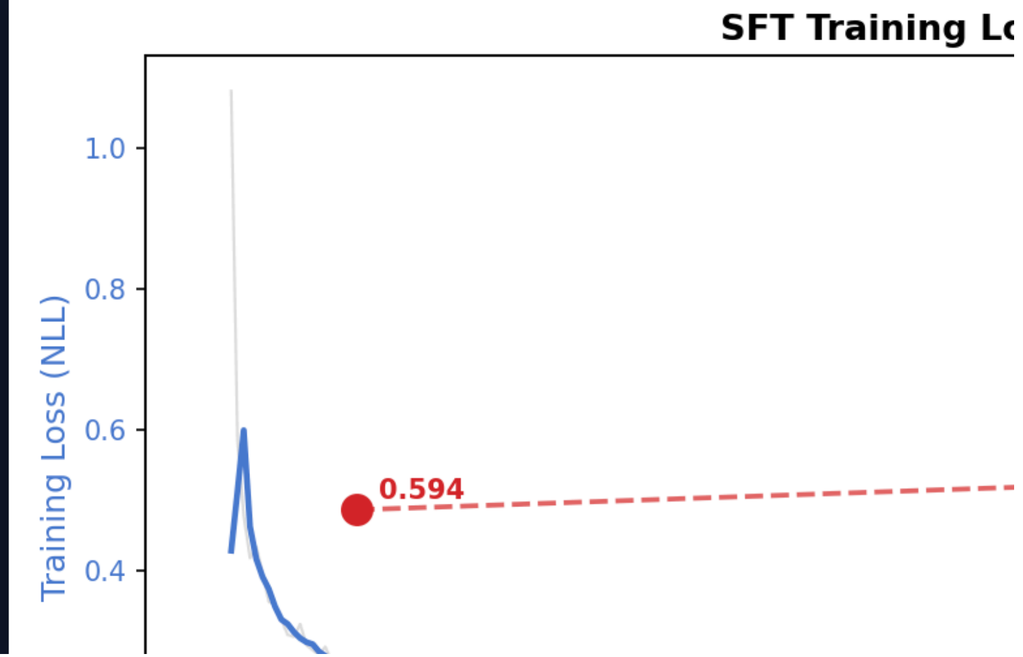

Trainin

Pipeline Overview



- **Goal:** improve Fermi-style estimation by fixing labels, grounding with facts, then fine-tuning
- **Data cleanup:** found and corrected incorrect ground truths in the final dataset before training
- **Grounding:** built a static fact file (≈500 questions)

SFT Training Dynamics



GRPO Reward Function

Reward formula (RL v2)

```
reward = 0.85 × oom_score
       + 0.15 × struct_score
```

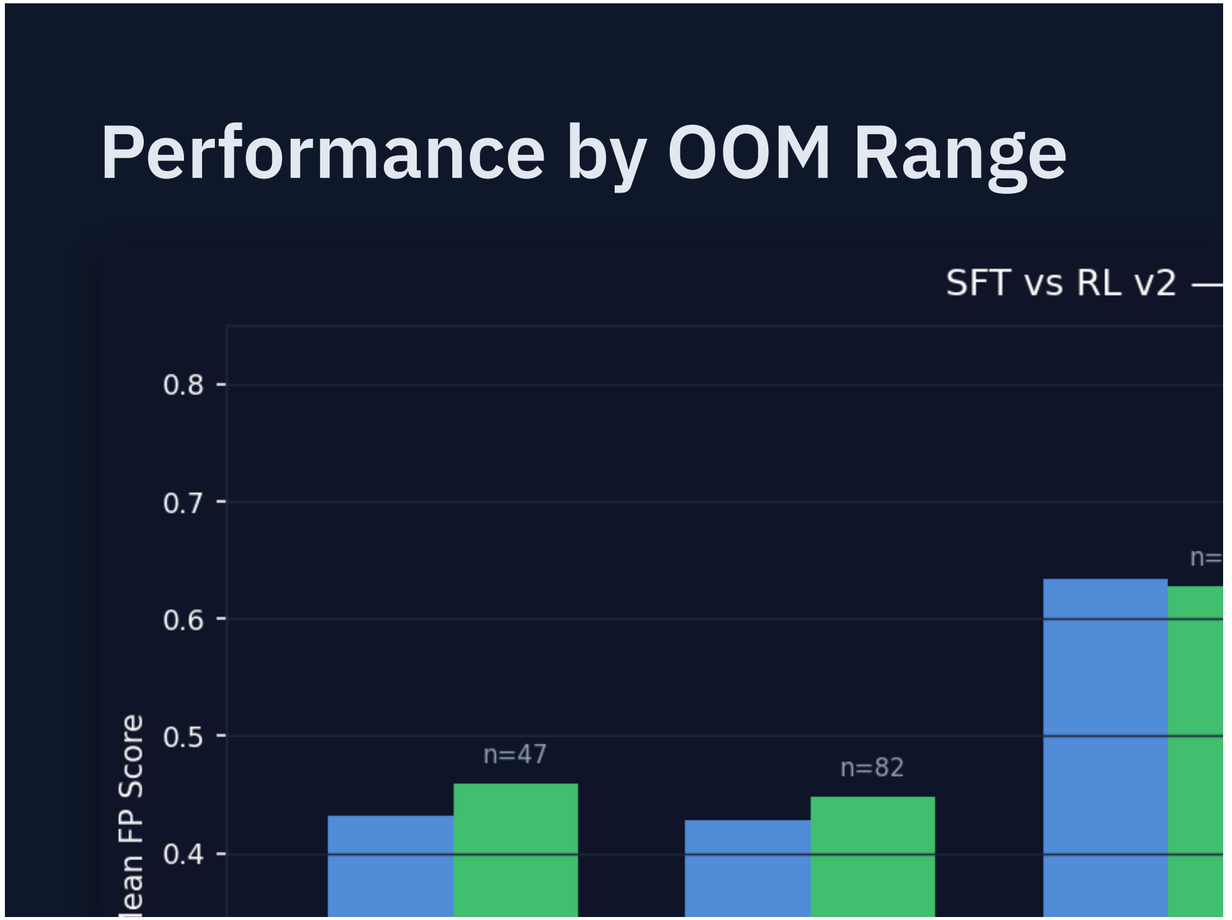
- `struct_score` = (has_factors + has_calc + has_formula) / 3
- Tail OOM multiplier: **×1.3** if gold OOM ≤ −4 or ≥ 10
- Parse fail → **−1.0** (hard penalty)


Re

Key Results — SynthFP-Free Eva

Baseline = Qwen3-4B-Instruct-2507, no fine-tuning ·

Metric
Mean score
Exact OOM accuracy
Within 1 decade
Within 2 decades
Parse OK
Score = 0 (complete fail)


Performance by OOM Range
SFT vs RL v2
0.8
0.7
0.6
0.5
0.4
Mean FP Score
n=47
n=82
n=

Qualitative Example — parsed re

SynthFP-free eval (`eval_no_synthfp`, index **107**, id `107`). Gold OOM
JSON from eval logs; baseline wrong OOM; RL exact.

> **Question:** How many gallons of gasoline are consu

Baseline (no FT) · predicted OOM **11** · FP **≈0.001** ·
`fp_log10_error=2.996`


```
{
  "question": "How many gallons of gasoline are consumed in the US every day?",
  "assumptions": [
    {
      "name": "daily_gasoline_consumption_per_capita",
      "value": 300,
      "unit": "gallons",
      "low": 200
```

Qualitative Example — what cha

Baseline failure mode

- **Bad anchor:** Labels **300 gallons** as **daily** per capita — confuses something closer to **annual** usage with **per day**, blowing total consumption to ~10^{11} gal/day → `oom: 11` vs gold **8**.
- **Calculation:** Still runs **population × rate**, but the rate slot is wrong by orders of magnitude


Conc

What Worked

SFT

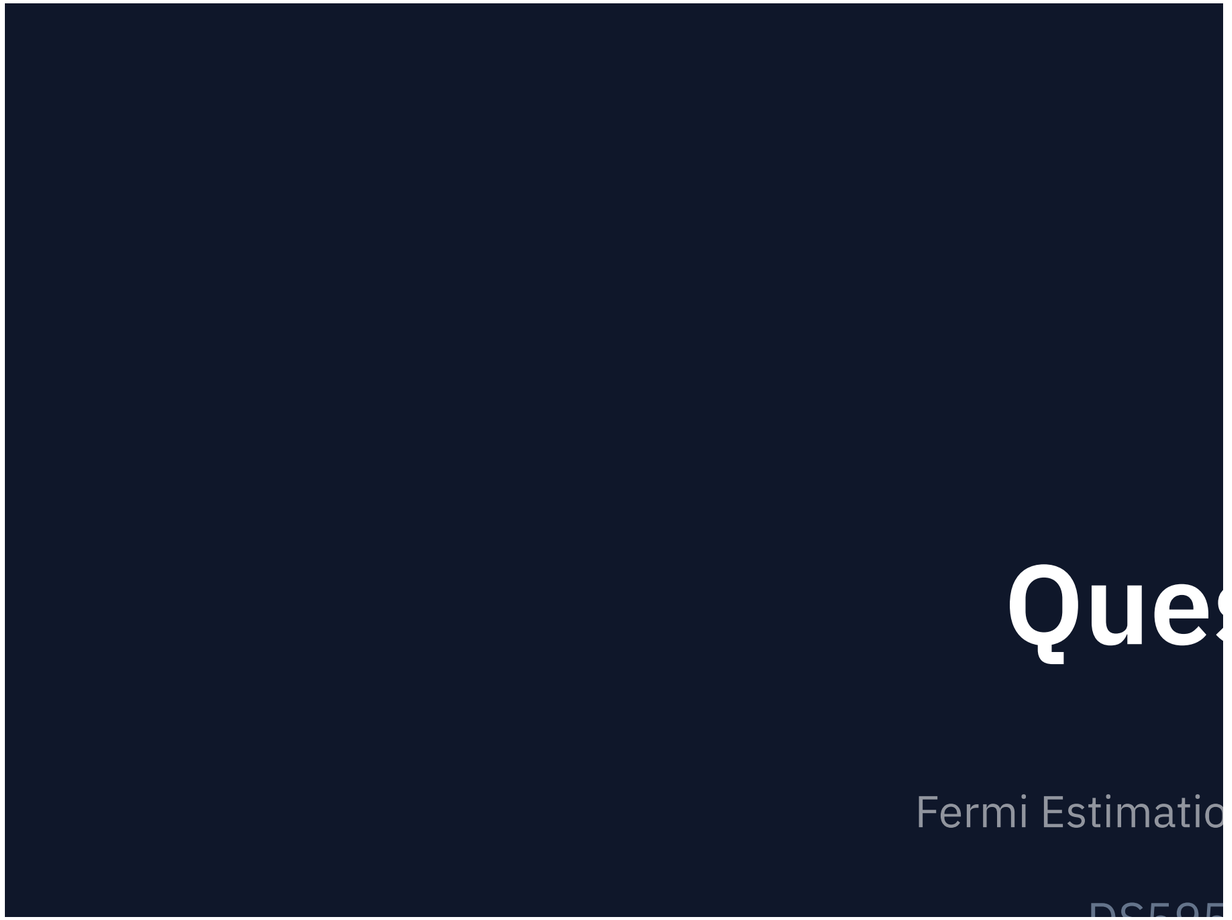
- +6.6% parse OK (92.2% → 98.8%)
- Structured reasoning format learned reliably
- Val score 0.518 → 0.691 over training

GRPO (RL v2)

What Didn't & Future Directions

What didn't work

- Tail OOM accuracy remains poor — high zero-score rate at extreme OOMs (≤−10, ≥20)
- Confidence is uncalibrated — equally wrong at high/medium confidence


Ques
Fermi Estimatio